

TECHNICAL SPECIFICATIONS I

FRAMEWORK CONTRACT FOR THE SUPPLY OF GEOSPATIAL PRODUCTS FOR THE COPERNICUS SECURITY SERVICE COMPONENT ON SUPPORT TO EU EXTERNAL AND SECURITY ACTIONS

SATCEN OP-05/26

APPENDIX A TO THE TENDER SPECIFICATIONS

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1 INTRODUCTION

This document provides the technical specifications for the elaboration of the geospatial products for the Copernicus Security Service component on Support to EU External and Security Actions (SESA). It defines the general technical and operational requirements governing the production, structure, content and delivery of the products to be supplied under the Contract.

The technical specifications applicable to this procurement procedure are divided into an open-access document and a controlled-access document. This open-access document contains the general specifications and the technical framework relevant to the procurement procedure, while the controlled-access document (Appendix B – Technical Specifications II) contains the detailed provisions relating to extraction criteria, topology, cartographic representation, output formats, quality checks, and validation.

In view of the operational sensitivity of part of the technical information, access to Appendix B shall only be granted by the Contracting Authority upon request, following the procedure stated in Annex I to the Invitation to Tender, and subject to the signature of a non-disclosure undertaking (Appendix K), to interested economic operators under equal conditions and sufficiently in advance of the deadline for submission of tenders.

The subsequent sections of this document are structured as follows:

- Section 2 provides a general description of the Product Types
- Section 3 defines the requirements related to Earth Observation imagery and collateral data used for production.
- Section 4 describes the production parameters that determine the scope of the Products and influence both pricing and production time (as further defined in Appendix D – Calculation of total price and production time of Orders).
- Section 5 outlines the general cartographic and analytical principles applicable to the production of the Products.
- Section 6 describes in detail the characteristics and requirements of each Product Type.
- Section 7 defines the service workflow governing the execution of Orders.
- Section 8 specifies the technical requirements applicable to delivery methods.

2 OVERVIEW OF PRODUCT TYPES

Under the FWC, the Contracting Authority may request the production of five (5) Product Types:

- **First Impression Report (FIR):** A rapid reporting and analysis product providing an initial assessment of the AOIs based on satellite imagery and relevant collateral sources. The purpose of the FIR is to support timely situational awareness, monitoring of activities and detection of significant changes within the AOI.
- **Briefing Note (BN):** A reporting and analysis product combining information derived from satellite imagery, collateral sources and previously available analyses. The BN provides contextualised information and interpretation of the AOI, including environmental and operational aspects.
- **Digital Geographic Information (DGI):** A product providing up-to-date cartographic and thematic information over a defined AOI. It includes detailed geospatial data and may incorporate map series depending on the size and complexity of the area.
- **Mapbook (MB):** A product combining cartographic outputs with analytical content related to activities within the AOI. It may include thematic mapping (e.g. population, socio-economic factors, terrain analysis) derived from GIS analysis and open-source data, and supports regional analysis, border monitoring and operational planning.
- **Country Map Coverage (CMC):** A single topographic map covering a region or a country, including essential vector data such as administrative boundaries, transportation networks, populated places and key infrastructure. The purpose of the CMC is to provide a rapid overview of the geographic and socio-economic context of the area.

The detailed technical specifications, production parameters and delivery requirements applicable to each Product Type are defined in the following sections of this document.

3 EO & GEOSPATIAL DATA ACCESS

3.1 Access to Earth Observation Data

The geospatial production under the Copernicus SESA Service primarily relies on the analysis of Very High Resolution (≤ 1 m) satellite imagery, mainly optical and/or SAR data. Depending on the analytical requirements of the Order, imagery at other spatial resolutions may also be used.

The spatial resolution of the EO data used for production shall determine the applicable Input Imagery Resolution Class (IIRC), as defined in Section 4.2.

3.1.1 Imagery provided under the ESA Rapid Response Desk (RRD)

EO imagery for the execution of the Contract is provided under the framework of the European Space Agency (ESA) Rapid Response Desk¹ (RRD), which supports authorised Copernicus users with access to data from the Copernicus Contributing Missions (CCM), including Very High Resolution (VHR) and High Resolution (HR) optical and SAR missions.

The Contracting Authority shall act as the sole interface with ESA for the purposes of imagery ordering under the RRD mechanism. For each activation:

- The identification, selection and ordering of imagery shall be performed by the Contracting Authority.
- The Contracting Authority shall provide the relevant EO imagery to the Contractor.
- The production of the Product shall be deemed activated upon provision of the imagery.

The Contractor shall be responsible for:

- Reviewing the latest information on available missions, technical specifications, and applicable data access conditions.
- Ensuring the capacity for rapid download, storage and processing of EO imagery.

Satellite data provided under the RRD mechanism shall be subject to the applicable ESA user licence and licensing conditions. Following signature of the Framework Contract, the Contractor shall sign the ESA user licence where required and comply with all applicable data usage conditions.

3.1.2 Use of Sentinel and other open-access EO data

Open and free EO data, including Sentinel missions, may be used where appropriate, either as primary input data or as collateral information.

Information about Sentinel missions is available through the Sentinel Online platform². Sentinel data are accessible via the Copernicus Data Space Ecosystem³ (CDSE) or other official Copernicus

¹ <https://www.rapidresponse.copernicus.eu/>

² <https://sentinels.copernicus.eu/web/sentinel/home>

³ <https://dataspace.copernicus.eu/>

distribution platforms, subject to the applicable Copernicus Sentinel Data Terms and Conditions in force at the time of access.

The Contractor shall ensure compliance with the applicable licensing terms and conditions governing any open-access EO data used in the execution of the Contract.

3.1.3 Use of other EO data sources

Where a Digital Elevation Model (DEM) is required for the execution of an Order, the Copernicus DEM shall constitute the primary elevation dataset. The use of other types of EO data (e.g. 3D models, high-resolution DEMs, interferometric products, night-time imagery or hyperspectral data) is not expected to be frequent. However, where such data is needed for the generation of a Product, they shall be used in accordance with the applicable licensing conditions.

During the set-up phase of the Contract, the necessary technical and administrative arrangements shall be established to ensure the appropriate use of EO imagery throughout the execution of the FWC.

3.2 Use of collateral geospatial data

In addition to the EO imagery provided under Section 3.1, the Contractor may use collateral and ancillary data, including open-sourced, where necessary to fulfil the requirements of the Order.

Such data may include (non-exhaustive list), as appropriate:

- Open-source geospatial datasets (e.g. OpenStreetMap, embassypages.com).
- Reference datasets supporting geospatial validation and contextual interpretation.
- Environmental or thematic datasets.
- Geotagged open-source information.
- Publicly available reports or other open-source information relevant to the analysis.

Where relevant to the execution of an Order and subject to availability and suitability, due consideration shall be given to data and products generated under other Copernicus Services and to local or national authoritative datasets from EU Member States, which may be accessible, among other means, through the Copernicus Reference Data Access (CORDA⁴) portal, subject to the applicable terms and conditions governing their access and use.

⁴ <https://corda.eea.europa.eu/>

4 PRODUCTION VARIABLES

The execution of each Order under the FWC are defined by a set of production variables that determine the scope, analytical depth and operational requirements of the requested Product. These variables are specified in the Order issued by the Contracting Authority and govern the applicable technical requirements, production effort and delivery conditions.

The production variables described in this Section define the technical and operational meaning of the parameters used for the execution of the Products. These variables are independent parameters and may be combined as appropriate to fulfil the objectives of the Order.

The economic and production time implications associated with these variables are defined in Appendix D - Calculation of the Total Price and Production Time of Orders, which establishes the methodology and correcting factors used to calculate the Total Price and Production Time applicable to each Order.

4.1 Area of Interest (AOI)

The AOI defines the geographical extent within which geospatial production shall be performed for a given Order. The AOI shall be specified in the Order issued by the Contracting Authority and shall be expressed in square kilometres (km²).

4.1.1 AOI definition in Orders

The AOI shall be provided in a clear and unambiguous manner, using one or more of the following (as applicable):

- AOI polygon (vector geometry).
- AOI defined by administrative unit(s) with sufficient identification.
- AOI defined by a buffer around a feature or corridor (with buffer distance).
- AOI defined by a grid / tiling reference.
- Any other spatial definition allowing the Contractor to delineate the AOI and quantify its area.

Where the production involves several Locations of Interest (LOIs) within the same activation, the Order may specify multiple AOIs (e.g. one AOI per LOI), provided that each AOI is uniquely identified.

4.1.2 Standard AOI per Product Type

For reference and standardization purposes, each Product Type is associated with a standard AOI size (or AOI definition), as summarized in Table 1.

Orders may request AOIs smaller or larger than the standard AOI, depending on operational needs. The technical and production implications of varying AOI sizes (including any impact on production time and/or pricing) are defined in Appendix D - Calculation of the Total Price and Production Time of Orders.

Product Type	Standard AOI
First Impression Report	100 km ²
Briefing Note	100 km ²
Digital Geographic Information	200 km ²
Mapbook	400 km ²
Country Map Coverage	Region / Country

Table 1 Standard AOI per product type

4.1.3 Practical implications of AOI size

The AOI size influences, inter alia:

- The number of images required to cover the AOI (including potential mosaicking).
- The number of map sheets/pages required (notably for Mapbooks and DGI map series).
- The extent of feature extraction and validation effort.

Where an AOI exceeds the standard AOI, the Contractor shall ensure production scalability while maintaining consistent cartographic quality and compliance with the applicable specifications.

4.2 Input Imagery Resolution Classes (IIRC)

The production of Products under the Copernicus SESA Service primarily relies on Very High Resolution (≤ 1 m) imagery but may also use other spatial resolutions depending on the analytical requirements of the Order.

For consistency, input imagery is categorised by IIRC. The IIRC classes below are used for technical scoping (and are referenced in Appendix D - Calculation of the Total Price and Production Time of Orders):

Input Imagery Resolution Class (IIRC)	Description
Input Imagery Resolution Class 1 (IIRC1)	Resolution ≤ 1 m
Input Imagery Resolution Class 2 (IIRC2)	1m < resolution ≤ 4 m
Input Imagery Resolution Class 3 (IIRC3)	4m < resolution ≤ 7 m
Input Imagery Resolution Class 4 (IIRC4)	7m < resolution ≤ 10 m
Input Imagery Resolution Class 5 (IIRC5)	10m < resolution ≤ 30 m
Input Imagery Resolution Class 6 (IIRC6)	30m < resolution ≤ 100 m
Input Imagery Resolution Class 7 (IIRC7)	100m < resolution ≤ 300 m
Input Imagery Resolution Class 8 (IIRC8)	Resolution > 300m

Table 2 Input Imagery Resolution Class

The use of input imagery resolution classes IIRC5 to IIRC8 is only foreseen in exceptional situations linked to specific products.

The applicable IIRC in each Order shall be specified.

4.2.1 Use of multiple IIRCs within the same Product

Certain Product types (FIR, BN) may require the use of imagery belonging to more than one IIRC within the same Order (e.g. combining HR and VHR imagery for complementary purposes).

Where multiple IIRCs are required, the Order shall specify the AOI distribution per IIRC (in km²).

4.2.2 Applicability of IIRCs by Product Type

The table below summarizes which IIRCs may be used by Product Type.

Product Type	IIRC1	IIRC2	IIRC3	IIRC4	IIRC5	IIRC6	IIRC7	IIRC8
FIR	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
BN	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
DGI	Yes	Yes	Yes	Yes	No	No	No	No
MB	Yes	Yes	Yes	Yes	No	No	No	No
CMC⁵	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Table 3 Product Types and IIRCs

The table below summarizes which Product Types envisage the use of multiple imagery from different IIRCs within one Product.

Product Type	Multiple IIRCs within the same product
FIR	Yes
BN	Yes
DGI	No
MB	No
CMC⁶	N/A

Table 4 Product types envisaged with more than one IIRC

4.3 Product category

Orders issued under the FWC may be requested under one of the following categories:

- New Product.
- Product Update.
- Product Upgrade.

The distinction between these categories reflects the degree to which previously generated data and products may be re-used and the extent of additional production effort required.

⁵ For CMC, satellite images (e.g. Sentinel, Open Source imagery) can be used as background information.

⁶ As specified in Section 6.5, the CMC product type is defined to allow for the use of intrinsically embedded VHR satellite imagery for the provision of annotated close-ups of highlighted features.

4.3.1 New Product

A New Product consists of the production of a deliverable based on the technical specifications defined in the Order, without re-use of a previous release under the same activation.

The production of a New Product requires the Contractor to perform all tasks necessary for the creation of the requested Product, including:

- Analysis and interpretation of the input imagery and relevant collateral information.
- Extraction, compilation and attribution of geospatial data in accordance with the applicable specifications.
- Preparation of the cartographic outputs and associated analytical content.

4.3.2 Product Update

A Product Update consists of the production of a new release of an existing Product based on newly acquired imagery, while maintaining the same:

- Product Type.
- Technical specifications and level of detail.
- Analytical scope.

A Product Update is typically requested to reflect changes over time within the same operational context.

The Contractor shall re-use the production of a previous release provided by the Contracting Authority and update the Product to represent the situation at the time of the new analysis. This includes:

- Analysing the newly acquired imagery.
- Identifying and documenting changes compared to the previous release.
- Updating the relevant geospatial data and cartographic outputs accordingly.

Where relevant, the updated release may include additional elements already defined within the existing product specifications (for example the inclusion of newly identified features such as roadblocks or infrastructure changes).

The Contractor shall ensure consistency between the updated release and the previous version.

4.3.3 Product Upgrade

A Product Upgrade consists of a modification of a previously delivered Product resulting from revised or enhanced technical or analytical requirements compared to the original Order.

A Product Upgrade reflects a change in scope or specifications, rather than a simple continuation of the original requirements.

The Contractor shall partially re-use the production of a previous release and complement or adapt the Product to meet the revised requirements. This may include, for example:

- Adaptation or population of additional attributes in the applicable data model.
- Inclusion of new thematic layers or analytical components not present in the previous release.
- Enhancement of the level of detail or analytical depth.
- Modification of the structure or content of the deliverable.

In order to produce a Product Upgrade, the Contractor shall be capable of:

- Performing all tasks necessary for the creation of the corresponding Product.
- Analysing the imagery and adapting the existing datasets to the revised specifications.
- Updating the cartographic outputs and associated analytical content.

4.3.4 Applicability

The applicable Product category (New Product, Product Update or Product Upgrade) shall be explicitly specified in the Order.

The economic and production time implications associated with each category are defined in Appendix D - Calculation of the Total Price and Production Time of Orders.

4.4 Production mode

Orders issued under the FWC may be requested either in Standard Production Mode or Urgent Production Mode. The applicable production mode shall be explicitly specified in the Order issued by the Contracting Authority.

Urgent Production Mode is intended to support operational situations where the requested Product must be delivered within a significantly reduced timeframe compared to standard production timelines.

4.4.1 Standard Production Mode

Standard Production Mode corresponds to the normal production timelines applicable to each Product Type under standard operating conditions.

Under Standard Production Mode, the Contractor shall perform all tasks required for the production of the Product in accordance with the technical specifications defined in this document and within the production time specified in the Order.

4.4.2 Urgent Production Mode

Urgent Production Mode applies where operational circumstances require accelerated delivery compared to Standard Production Mode.

Urgent Production Mode does not modify the technical specifications, analytical requirements or quality standards applicable to the requested Product. The Contractor shall ensure that the level of quality and analytical depth remains consistent with the requirements defined for the relevant Product Type.

The use of Urgent Production Mode reflects the need to perform the same production activities within a reduced timeframe and may therefore require additional production effort and resource mobilisation.

4.4.3 Applicability

Urgent Production Mode may be requested for the Product Types indicated in the table below.

Product Type	Urgent Mode
FIR	Yes
BN	Yes
DGI	Yes
MB	Yes
CMC	No

Table 5 Product types and urgent production

The implications of Urgent Production Mode on price and production time are defined in Appendix D - Calculation of the Total Price and Production Time of Orders.

4.5 Temporal complexity (TC)

Products may rely on one or more imagery acquisition dates to support interpretation and change assessment. Two Temporal analysis categories are defined:

- Standard temporal complexity.
- Augmented temporal complexity.

4.5.1 Standard Temporal Complexity

Corresponds to analysis based on a single acquisition date or limited multi-date comparison, where the temporal dimension does not require significant additional analytical effort. This includes visual comparison of one to five (5) coverages⁷ without systematic temporal processing.

4.5.2 Augmented Temporal Complexity

Corresponds to analysis where the temporal dimension introduces significant additional analytical effort, mainly due to the number of acquisitions and/or the temporal period covered, requiring detailed interpretation and comparison of multiple acquisition dates over extended periods. It does not, by itself, imply the application of specialised multi-temporal processing or advanced analytical methodologies.

4.5.3 Applicability

The distinction between Standard Temporal Complexity and Augmented Temporal Complexity shall be determined based on the additional analytical effort resulting from the number of acquisition dates and the temporal period covered. Augmented Temporal Complexity shall apply where the analysis involves more than five (5) coverages of the same AOI, corresponding to different acquisition dates, and where detailed multi-date interpretation over an extended temporal period is required.

Temporal Complexity reflects the additional effort required to analyse and interpret multiple acquisitions over time within the same area. It relates to the number of acquisition dates and the

⁷ In this context, 'coverage' shall be understood as one complete spatial coverage of the relevant AOI for a given similar reference date, irrespective of the number of individual images to achieve such coverage.

temporal extent of the analysis and does not determine the analytical methodology or processing technique to be applied.

The applicable TC category shall be specified in the Order by the Contracting Authority. The economic and production time implications associated with each TC category are defined in Appendix D - Calculation of the Total Price and Production Time of Orders.

Product Type	Standard TC	Augmented TC
FIR	Yes	Yes
BN	Yes	Yes
DGI	Yes	No
MB	Yes	No
CMC	No	No

Table 6 Product Types and TC

4.6 Advanced analysis

Orders issued under the FWC may require the performance of Advanced Analysis in addition to the standard production of a Product.

Advanced Analysis consists of analytical activities that exceed standard feature extraction, routine interpretation or cartographic production, and require the application of structured geospatial or analytical methodologies tailored to the operational objective of the Order. It may involve the integration of additional datasets, the use of specialised geospatial processing techniques, or the implementation of analytical workflows requiring enhanced methodological design.

Advanced Analysis may be requested under one of the levels described below, and the applicable level of complexity shall be explicitly specified in the Order issued by the Contracting Authority.

4.6.1 Advanced Analysis

Advanced Analysis involves analytical tasks performed using established tools and methodologies available within standard GIS and remote sensing software environments.

Such tasks typically require the acquisition, preparation and integration of specific datasets, and the configuration and controlled use of processing techniques by qualified operators to ensure reliable and accurate outputs. The processing may be partially or largely automated but requires an appropriate level of technical expertise to select methods, adjust parameters and validate results.

Advanced Analysis may include, for example:

- **Network analysis:** Creation, preparation and optimisation of a road network dataset used for road network analysis. Elements such as intersections or road characteristics (e.g. speed limit, functional class, lane width and number of lanes) shall be correctly defined in order to support common networking scenarios such as best route, closest facility, travelling salesman or service areas.

- Visibility Analysis: Assessment of observation capabilities and terrain visibility using key terrain, observation posts or other relevant locations.
- Spatial Analysis: Creation, query, mapping and analysis of raster data; integrated raster/vector analysis; derivation of new information from existing datasets; querying across multiple data layers.
- Supervised and unsupervised classification of imagery: For instance, Land Use / Land Cover classification, vegetation density levels or flooded areas.
- SAR data analysis, including amplitude change detection, coherence change detection and multi-temporal coherence analysis.
- Calculation of spectral indices.
- Spatial multi-criteria analysis.
- Multi-source data fusion.

This category does not require the design or implementation of fully customised analytical methodologies, which are characteristic of High Complexity Advanced Analysis.

4.6.2 High Complexity Advanced Analysis

High Complexity Advanced Analysis requires the design and implementation of dedicated methodologies to fulfil specific technical or analytical requirements. This category typically requires the use of advanced EO techniques, specialised expertise and structured analytical workflows involving multiple processing steps and validation procedures.

High Complexity Advanced Analysis may include, for example:

- Advanced spatial modelling (e.g. hydrological modelling).
- Advanced time-series analysis requiring dedicated methodological design, including trend analysis, temporal modelling, anomaly detection or other customised analysis of long or dense EO time series.
- Interferometric and/or polarimetric SAR time-series processing and analysis.
- Time-series analysis, including interferometric and/or polarimetric SAR data processing and analysis.
- AIS data analysis.
- Generation of on-demand high-resolution DEMs.
- Hyperspectral data analysis.
- Advanced 3D analytics.
- LiDAR data analysis.
- Modelling and mapping the spatial and temporal development of night-time lights.
- Geostatistical analysis and modelling of spatial or spatio-temporal phenomena.

The layout, content and methodology of High Complexity Advanced Analysis shall be defined on a case-by-case basis in the Order.

4.6.3 Applicability

Advanced Analysis may be requested in combination with:

- Any applicable Product Type except CMC.
- Standard or Urgent Production Mode.
- Any Temporal Complexity category.

The applicable Advanced Analysis level shall be explicitly specified in the Order.

The implications of Advanced Analysis for price and production time are defined in Appendix D – Calculation of the Total Price and Production Time of Orders.

Product Type	No Advanced Analysis	Advanced Analysis	High Complexity Advanced Analysis
FIR	Yes	Yes	Yes
BN	Yes	Yes	Yes
DGI	Yes	Yes	Yes
MB	Yes	Yes	Yes
CMC	Yes	No	No

Table 7 Product types and advanced analysis categories

4.7 Production time

Production Time represents the time required for the Contractor to produce a Product under the conditions specified in the Order.

The Production Time applicable to each Order shall be determined based on the production variables defined in this Section and shall reflect the production effort required to fulfil the requested Product under the specified conditions.

For reference, each Product Type is associated with a Standard Production Time corresponding to a baseline production scenario defined as:

- New Product.
- Standard AOI.
- IIRC1.
- Standard Production Mode.
- Standard Temporal Complexity.
- No Advanced Analysis.

The Standard Production Time for each Product Type is defined in Section 6.

The Production Time applicable to a specific Order may vary depending on the combination of production variables specified in the Order, including but not limited to AOI, IIRC, Product category, Production mode, Temporal Complexity and Advanced Analysis.

The methodology used to calculate the Production Time applicable to each Order, including any applicable correcting factors, is defined in Appendix D - Calculation of the Total Price and Production Time of Orders.

5 GENERAL CARTOGRAPHIC AND ANALYTICAL PRINCIPLES

The Contractor shall ensure that the analytical approach and cartographic representation of each Product are adapted to the specific operational scenario and user requirements defined in the Order. The content, level of detail and analytical outputs shall be tailored as necessary to ensure that the information provided is relevant, accurate and fit for purpose.

Where standard cartographic symbols, conventions or predefined representations are not sufficient to adequately convey the analytical content, the Contractor shall design and apply appropriate dedicated symbology or visual elements. Such adaptations shall ensure clarity, consistency and ease of interpretation for the end user, while maintaining coherence with the overall product specifications.

The detailed technical provisions governing cartographic and analytical production (including extraction criteria, topology rules, cartographic representation standards, output formats, and quality check and validation requirements) are set out in Appendix B - Technical Specifications II.

6 PRODUCT TYPES

As stated in Section 2, this FWC covers the on-demand production of five (5) Product Types, described in the following sub-sections:

- First Impression Report (FIR).
- Briefing Note (BN).
- Digital Geographic Information (DGI).
- Mapbook (MB).
- Country Map Coverage (CMC).

Each Product Type is designed to address specific operational and analytical needs within the Copernicus SESA Service. Examples of use cases illustrating the range and flexibility of the Copernicus SESA Service are available in the SESA website⁸. Such examples are indicative and do not limit the possible scope of Orders issued under the FWC.

For each Product Type, the following information is provided:

- **Product Description:** Purpose, main characteristics and intended application.
- **Content:** Description of the information and elements to be included.
- **Standard AOI:** Indicative spatial coverage, as indicated in Section 4.1.2. Orders can request a different area, either larger or smaller than this value.
- **Standard Production Time:** Maximum efficient production time under standard conditions, expressed in working days. The service workflow is described in detail in Section 7.
- **Estimated Number of Pages:** Indicative maximum number of pages, including an indication of the typical length. Cover pages and blank pages shall not be counted.
- **Page Size:** Expected page format for the Product, where applicable.
- **Deliverable Format:** Expected output structure and format.

The characteristics described for each Product Type reflect standard conditions. The actual scope, AOI, analytical depth, imagery characteristics, production mode, production time and any additional requirements shall be explicitly specified in the Order and shall be governed by the production variables defined in Section 4.

⁸ <https://sesa.security.copernicus.eu/>

6.1 First Impression Report (FIR)

Product Description	FIR is a rapid analytical reporting product providing an initial assessment of a Location of Interest (LOI) based on satellite imagery and relevant collateral sources. The purpose of the FIR is to support timely decision-making through rapid situational awareness, monitoring of activities and detection of significant changes within the specified LOI.
Content	The FIR may include, as appropriate: • Satellite imagery. • Basic reference information. • Comprehensive and interpretative intelligence analysis (in the form of charts and tables). • Relevant collateral information (e.g. photographs or open-source elements, AIS data). • Surrounding areas in overview/reference maps may be requested using collateral geospatial information, including open-source data such as OpenStreetMap. The content shall be tailored to the operational objective specified in the Order.
Standard Area of Interest	100 km ²
Standard Production Time	3 working days
Estimated Number of Pages	From 4 to 10 pages
Page Size	A3 - A4
Input Imagery Resolution Class	☒ IIRC1 ☒ IIRC2 ☒ IIRC3 ☒ IIRC4 ☒ IIRC5 ☒ IIRC6 ☒ IIRC7 ☒ IIRC8 More than one IIRC may apply within the same activation.
Product Update / Upgrade	☒ New Product ☒ Product Upgrade ☒ Product Update
Urgent Production	☒ Standard Mode ☒ Urgent Mode
Temporal Complexity	☒ Standard Temporal Complexity ☒ Augmented Temporal Complexity

Advanced Analysis	☒ No Advanced Analysis ☒ Advanced Analysis ☒ High Complexity Advanced Analysis
Examples of Use Cases	The list supplied below represents some common examples of use cases (non-exhaustive list): • Activity Report. • Camp Analysis. • Environmental compliance monitoring. • Infrastructure monitoring. • Large-area monitoring over open-water and object detection

6.2 Briefing Note (BN)

Product Description	BN is a fast reporting and analysis product. It is produced using information from satellite imagery and collateral sources, and aggregated findings of previous analysis. The report includes description and analysis of the context and environment of the AOI.
Content	The BN may include, as appropriate: • Satellite imagery. • Basic reference information. • Comprehensive and interpretative intelligence analysis (in the form of charts and tables). • Relevant collateral information (e.g. photographs or open-source elements). • Surrounding areas in overview/reference maps may be requested using collateral geospatial information, including open-source data such as OpenStreetMap. The content shall be tailored to the operational objective specified in the Order.
Standard Area of Interest	100 km ²
Standard Production Time	10 working days
Estimated Number of Pages	From 6 to 20 pages
Page Size	A3 - A4
Input Imagery Resolution Class	☒ IIRC1 ☒ IIRC2 ☒ IIRC3 ☒ IIRC4 ☒ IIRC5 ☒ IIRC6 ☒ IIRC7 ☒ IIRC8 More than one IIRC may apply within the same activation.
Product Update / Upgrade	☒ New Product ☒ Product Upgrade ☒ Product Update
Urgent Production	☒ Standard Mode ☒ Urgent Mode
Temporal Complexity	☒ Standard Temporal Complexity ☒ Augmented Temporal Complexity

Advanced Analysis	☒ No Advanced Analysis ☒ Advanced Analysis ☒ High Complexity Advanced Analysis
Examples of Use Cases	The list supplied below represents some common examples of use cases (non-exhaustive list): • Activity Report. • Camp Analysis. • Conflict Damage Assessment. • Non-EU Border Map.

6.3 Digital Geographic Information (DGI)

Product Description	DGI product provides up-to-date relevant cartographic and thematic information over a specific AOI. It includes the necessary information to determine the key geographic characteristics of an AOI. This product may also include a detailed map series, depending on the area and the distribution of data.
Content	The DGI may include, as appropriate: • Satellite imagery. • Extensive extraction of all feature classes (FC) present in the AOI. • Close-ups of highlighted features with annotations and collateral information (in the form of images, charts and tables). The content shall be tailored to the operational objective specified in the Order.
Standard Area of Interest	200 km ²
Standard Production Time	12 working days
Estimated Number of Pages	The number of pages is not limited (a representative DGI could contain 1-10 pages for map series). However, the product may include: • Overview reference image map. • Index map. • Detailed reference image map. • Detailed reference map series (final number depends on the area). • Detailed map of each critical infrastructure.
Page Size	A0 – A1 – A2 – A3
Input Imagery Resolution Class	☒ IIRC1 ☒ IIRC2 ☒ IIRC3 ☒ IIRC4 ☐ IIRC5 ☐ IIRC6 ☐ IIRC7 ☐ IIRC8
Product Update / Upgrade	☒ New Product ☒ Product Upgrade ☒ Product Update
Urgent Production	☒ Standard Mode ☒ Urgent Mode

Temporal Complexity	☒ Standard Temporal Complexity ☐ Augmented Temporal Complexity
Advanced Analysis	☒ No Advanced Analysis ☒ Advanced Analysis ☒ High Complexity Advanced Analysis
Examples of Use Cases	The list supplied below represents some common examples of use cases (non-exhaustive list): • Crisis Situation Picture. • Critical Infrastructure Analysis. • Reference Map. • Road Network Status Assessment. • Support to Evacuation Plan.

6.4 Mapbook (MB)

Product Description	MB is a product combining cartographic information with analysis of activity. The MB may be complemented with thematic maps (e.g. population, socioeconomic factors, terrain analysis, location of public services) based on GIS analysis and open-source data. This product type is useful for regional analysis, border monitoring or the design of security plans over a city, facility or event location. It includes basic vector information, mainly focused on the transport network, hydrology and hypsography. The imagery intelligence analysis will focus on the detailed description of relevant places or event locations within the AOI.
Content	The MB may include, as appropriate: • Descriptive introduction. • Satellite imagery. • Extraction of the main FCs present in the AOI. • Close-ups of highlighted features with annotations. • Relevant collateral information (in the form of images, charts and tables). • Comprehensive and contextual intelligence analysis. • Surrounding areas in overview/reference maps may be requested using collateral geospatial information, including open-source data such as OpenStreetMap. The content shall be tailored to the operational objective specified in the Order.
Standard Area of Interest	400 km ²
Standard Production Time	25 working days
Estimated Number of Pages	The number of pages is not limited (a representative MB could contain 10-50 pages for map series and 5-20 pages for text). However, the product shall include the following pages: • Overview reference image map. • Index map. • Thematic maps (optional). • Detailed reference map series (final number of pages will depend on the area and the representation scale). • Activity analysis maps (final number of pages will depend on the area and representation scale).
Page Size	A3

Input Imagery Resolution Class	☒ IIRC1 ☒ IIRC2 ☒ IIRC3 ☒ IIRC4 ☐ IIRC5 ☐ IIRC6 ☐ IIRC7 ☐ IIRC8
Product Update / Upgrade	☒ New Product ☒ Product Upgrade ☒ Product Update
Urgent Production	☒ Standard Mode ☒ Urgent Mode
Temporal Complexity	☒ Standard Temporal Complexity ☐ Augmented Temporal Complexity
Advanced Analysis	☒ No Advanced Analysis ☒ Advanced Analysis ☒ High Complexity Advanced Analysis
Examples of Use Cases	The list supplied below represents some common examples of use cases (non-exhaustive list): • Crisis Situation Picture. • Critical Infrastructure Analysis. • Reference Map. • Road Network Status Assessment. • Support to Evacuation Plan.

6.5 Country Map Coverage (CMC)

Product Description	CMC is a single topographic map covering a region or a country. It includes vector data such as administrative, transportation, population-related features, border crossing points and toponyms. The purpose of the CMC is to quickly inform the user about the administrative organisation and main transportation network of the area, as well as the socioeconomic context.
Content	The CMC includes the following information: • Descriptive introduction. • Satellite imagery. • Very basic thematic information (e.g. administrative units, elevation, major cities, main airports and harbors). • Close-ups of highlighted features with annotations (e.g. pictures, VHR imagery focusing on relevant features like airports or border crossing points). • Collateral information (in the form of pictures, charts and tables).
Standard Area of Interest	Region / Country
Standard Production Time	6 working days
Estimated Number of Pages	One map presenting the overview reference map.
Page Size	A0 - A1 - A2
Input Imagery Resolution Class⁹	☐ IIRC1 ☐ IIRC2 ☐ IIRC3 ☐ IIRC4 ☐ IIRC5 ☐ IIRC6 ☐ IIRC7 ☐ IIRC8
Product Update / Upgrade	☒ New Product ☒ Product Upgrade ☒ Product Update
Urgent Production	☒ Standard Mode ☐ Urgent Mode

⁹ For CMC, satellite images (e.g. Sentinel, open source imagery) can be used as background information. In addition, VHR satellite images may be used for the close-ups of highlighted features with annotations

Temporal Complexity	☐ Standard Temporal Complexity ☐ Augmented Temporal Complexity
Advanced Analysis	☒ No Advanced Analysis (N/A) ☐ Advanced Analysis ☐ High Complexity Advanced Analysis
Examples of Use Cases	The list supplied below represents some common examples of use cases (non-exhaustive list): • Crisis Situation Picture. • Critical Infrastructure Analysis. • Reference Map.

7 SERVICE WORKFLOW

The execution of each Order shall follow a structured workflow ensuring timely delivery¹⁰, quality control and validation of the Products. In the following paragraphs, the steps of the geospatial production will be further defined; additionally, a detailed workflow is shown in Figure 1, where the following steps will be included:

1. **Placement and acknowledgment of an Order.** Upon receipt of an Order from the Contracting Authority, the Contractor shall:
 - Acknowledge receipt within a maximum of four (4) hours, unless otherwise specified.
 - Analyze the Order and initiate internal coordination.
 - Sign and return the Order within two (2) working days.
2. **Start of Production.** Production shall start when the Contractor receives the EO imagery required for the execution of the Order. This moment shall define the **production start time** (T_0). The Contractor shall acknowledge receipt of the EO imagery for the activation within a maximum of thirty (30) minutes, unless otherwise specified.

To ensure that sufficient time remains within the overall production time to accommodate validation cycles and final acceptance, the Contracting Authority may indicate as well at this stage an initial **first delivery date**. This indicative intermediate milestone is intended to support the validation process by allowing an early submission of the Product for review. Where such a date is provided, the Contractor is expected to make reasonable efforts to submit a first version of the Product accordingly. However, this intermediate milestone shall not constitute a separate contractual deadline and shall not give rise to penalties if not met. Based on operational experience, late submission of the first version may reduce the time available for validation and corrective iterations, which may increase the risk of exceeding the Production Time specified in the Order and/or affect the overall quality of the final Product.

3. **Processing of EO data.** The Contractor shall process the EO imagery received from the Contracting Authority as part of the production workflow. The Contractor is expected to make processed imagery available to the Contracting Authority progressively, as it becomes available during production.

The progressive delivery of processed imagery does not constitute a formal delivery of the Product and shall not be considered as part of the validation process. This practice shall not create additional contractual obligations in terms of delivery deadlines or acceptance criteria.

4. **Product generation and internal quality control:** The Contractor shall:
 - Perform all production tasks required for the generation of the Product.
 - Ensure internal quality control prior to delivery.
 - Guarantee consistency and homogeneity across outputs.

Before delivery, the Product must comply with all applicable technical specifications.

¹⁰ A Product shall be considered formally delivered only when it has been made available through the agreed delivery method.

5. **Product delivery and validation.** The Contractor shall deliver the Product within the timeframe specified in the Order.

Upon delivery:

- The Contracting Authority shall perform validation.
- Validation time shall not be considered part of the Production Time.

If the Product is not accepted:

- The Contractor shall implement the required corrections.
- The corrected Product shall be resubmitted within the remaining Production Time.

6. **Acceptance and production closure.** Production will be considered closed when the Product is accepted by the Contracting Authority without further changes requested.

This moment shall define the **production end time**. The Contracting Authority shall notify the Contractor of such acceptance (**product acceptance**).

The **measured production time** shall be calculated as the period between the production start time and the production end time and shall be equal to or less than the production time specified in the Order. If the measured production time exceeds the time specified in the Order, penalties may be applied to the Contractor, as detailed in Section II.15 of Annex II – Draft Framework Contract.

7. **Post-Production Activities:** Following Product acceptance:
- The Contractor shall submit the Technical Report within five (5) working days.
 - The Contracting Authority shall perform verification and acceptance of the Technical Report.
 - The Contractor may proceed with invoicing in accordance with Annex II.

The number of Orders may vary depending on operational demand. The Contractor shall maintain sufficient production capacity to manage multiple Products in parallel, in line with the operational requirements of the Service.

As each product has a different production time, the maximum production capacity is determined by the number of products to be produced in parallel, rather than by the number of orders placed simultaneously. Based on operational experience, peak demand scenarios may involve a high number of Products being produced in parallel (up to approximately 20 Products, assuming 50% of such products will be fast reporting products, such as FIRs). This estimate is indicative and non-binding and shall not be interpreted as a limitation or guaranteed maximum workload. In case of reaching or exceeding this estimated maximum number of products, Satcen will make its best efforts to define the priorities and assist the contractor in reorganizing its workload.

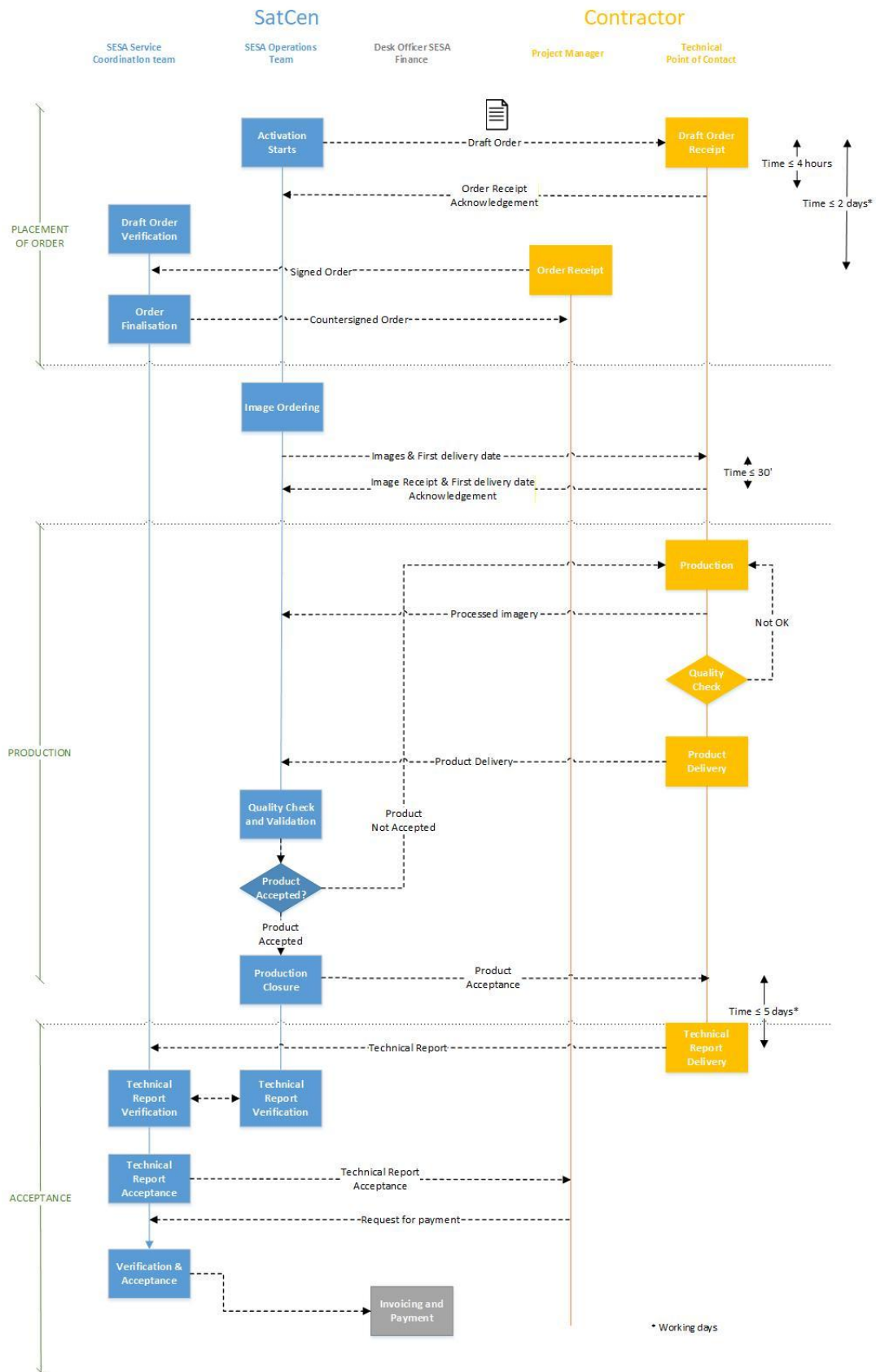


Figure 1 Production workflow

8 DELIVERY METHODS

The delivery of Products shall be performed using the mechanisms defined in this Section.

A Product shall be considered formally delivered when it has been made available through the delivery method specified and complies with the required output formats and specifications.

The contractor shall ensure that all data (including vector and raster data, input and processed imagery and quality control reports) are made available for publication through one of the following mechanisms, as indicated in Section 4.6.2 in Annex I – Tender Specifications:

1. Primary delivery method

A SFTP server, to be provided, managed and maintained by the Contractor.

The SFTP server may be hosted on the Contractor's premises or in a secure third-party data center. In all cases, the Contractor shall ensure an adequate level of information security and protection against unauthorized access.

2. Alternative delivery method

The alternative solution shall provide equivalent levels of security, reliability and data integrity. (e.g. secure drop zone or equivalent platform).